

# SEARCHING

Q7)

## Question:

| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Session | Searching | Question Information | Level 1 | Challenge 7 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|----------------------|---------|-------------|
| <p>Question Description:</p> <p>Suresh have "N" rectangles.</p> <p>A rectangle is Silver if the ratio of its sides is in between [1.6, 1.7], both inclusive. Your task is to find the number of silver rectangles.</p> <p>Constraints:</p> <p><math>1 \leq N \leq 10^5</math></p> <p><math>1 \leq W,</math></p> <p><math>H \leq 10^9</math></p> <p>Input Format:</p> <p>First line: Integer "N" denoting the number of rectangles Each of the "N" following lines:</p> <p>Two integers W, H denoting the width and height of a rectangle</p> <p>Output Format:</p> <p>Print the output in a single line contains find the number of Silver rectangles.</p> |         |           |                      |         |             |

## PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int n,w,h,i,a,b,count=0;
5
6      scanf("%d",&n);
7
8      for(i=0;i<n;i++) {
9          scanf("%d%d",&w,&h);
10
11         if(w<h) {
12             a=w;
13             b=h;
14         } else {
15             a=h;
16             b=w;
17         }
18
19         if(16*a<=10*b && 10*b<=17*a)
20             count++;
21     }
22
23     printf("%d\n",count);
24
25     return 0;
26 }
```

## OUTPUT:

### TEST CASE 1:

```
4
185 100
180 100
170 100
160 100
2
```

### TEST CASE 2:

```
6
185 100
180 100
170 100
160 100
165 90
164 95
2
```